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Accounting for Changing That the Measuring Sensor Makes into the Frequency Characteristics of Brain Tissue Impedance

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Abstract:
Impedance spectroscopy is now widely used in various fields, including medicine. This method is employed to assess organ viability, diagnose diseases, and detect malignant neoplasms. Moreover, impedance measurement serves not only as a diagnostic tool but also as method for precise tumor localization before surgical intervention (e.g., during biopsy) and directly during surgical procedures. Despite promising results obtained in a number of studies, the method has not gained widespread adoption. The main obstacle is the need for medical certification of new technologies, which significantly limits opportunities for in-depth in vivo research. To simplify access to such studies, the use of a bipolar coagulation forceps is proposed. Unlike the standard four-wire measurement setup, this tool requires only two wires, which, however, may introduce measurement errors. The article details the methodology for conducting experiments on a sample with parameters similar to those of brain tissue. Results are presented for both the standard four-wire measurement model and the two-wire model corresponding to the bipolar coagulation forceps. It is shown that the values differ by up to 10%, depending on the frequency range. At frequencies above 100 kHz, the two-wire model exhibits significant deviations from the standard measurement model.

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